Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_28

1. Summary of Key Deviations

The analysis of Run_28 reveals a complex signature where a critical sensor failure is happening concurrently with a separate, physical system anomaly.

- **p_FMU (Critical):** The pressure sensor at the Fuel Metering Unit shows a **major negative drift** beginning in state s2, which worsens to a **complete signal dropout (-99.4%)** in state s5. This is a clear and critical sensor failure.
- **Flow_Bypass (Moderate):** A significant positive **offset** of up to **+18.8%** is observed, indicating a large amount of flow is being diverted.
- **p_Combustion (Minor):** A consistent, minor positive **offset** of **+8%** is present across all active states.
- **Flow_Engine1-4 (Minor):** All four engine flows show a persistent **negative offset** of **-3% to -5%** during the throttle-up phases (s4-s6).
- **Flow_Pump (Minor):** The pump flow shows a slight positive **offset** of around +6%.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**p_FMU Sensor & FMU/Injectors**	**Dual Fault: Sensor Dropout & Physical Blockage**	This is the only hypothesis that fits all the data. There are two independent faults: 1) The **p_FMU sensor has failed**, causing the dropout. 2) A **physical blockage exists downstream of the pump** (likely at the FMU or injectors). This blockage explains the low engine flow, high bypass flow, and high combustion pressure. The sensor failure is a separate, concurrent event.

2	**p_FMU Sensor**	**Sensor Dropout / Total Failure**	This explains the critical -99.4% reading. However, a sensor-only fault **cannot** explain the other physical symptoms: the low engine flows, high bypass flow, and high combustion pressure are all indicative of a real physical blockage.
3	**Fuel Metering Unit (FMU) / Injector Nozzles**	**Partial Blockage**	This explains the physically consistent pattern of low engine flow, high bypass flow, and high combustion pressure. However, it does *not* explain why the `p_FMU` sensor, specifically, drops out completely while other pressure sensors do not.
4	**Catastrophic Leak at FMU**	**Major Leak in FMU Casing/Piping**	A massive leak *at* the FMU could theoretically cause the local pressure (`p_FMU`) to drop to zero. However, this would also starve the engines of fuel, causing `Flow_Engine` to drop, and it would not explain why the upstream `p_Combustion` pressure is *high*. This hypothesis is inconsistent with the data.
5	**Multiple Sensor Failures**	**p_FMU Dropout & Other Sensor Errors**	It is extremely unlikely that the `p_FMU` sensor would fail simultaneously with multiple other sensors (`Flow_Engine` reading low, `Flow_Bypass` reading high) in a way that perfectly mimics a physical blockage. The dual-fault physical/sensor scenario is far more probable.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Failure of the p_FMU Pressure Sensor AND (2) a concurrent, separate physical blockage at the FMU or Injector Nozzles.**

Reasoning: The data clearly shows two distinct and unrelated problems happening at once. The first is the undeniable dropout of the `p_FMU` sensor. The second is a classic downstream blockage signature: engine flows are low while bypass flow and upstream pressure are high. Neither fault can explain the symptoms of the other. A `p_FMU` sensor failure doesn't cause a physical blockage, and a physical blockage doesn't cause a single, isolated pressure sensor to drop to zero while others read high. Therefore, the only logical conclusion is that the system has suffered two simultaneous, independent failures.